Global Food Price Monitor API (1.0.0)

Download OpenAPI specification: Download

API for monitoring global food prices using WFP data

health

Health Check

Responses

> **200** Successful Response

GET /health

Response samples

200

Content type
application/json

Copy

```
{
    "property1": "string",
    "property2": "string"
}
```

auth

Register a new user

Create a new user account. Username must be 3–50 characters; password must be at least 8 characters.

REQUEST BODY SCHEMA: application/json
required

`username` required	string (Username) [3 .. 50] characters
`email` required	string <email> (Email)
`password` required	string (Password) >= 8 characters

Responses

> **201** The newly created user

— **400** Username or email already registered

> **422** Validation Error

POST `/api/v1/auth/register`

Request samples

Payload

Content type
application/json

Copy

```
{
    "email": "jane@example.com",
    "password": "securepassword123",
    "username": "jane_doe"
}
```

Response samples

201 422

Content type
application/json

Copy

```
{
    "created_at": "2024-01-15T10:30:00Z",
    "email": "jane@example.com",
    "id": 1,
    "is_active": true,
    "username": "jane_doe"
}
```

Log in and obtain a JWT

Authenticate with username and password (OAuth2 password flow). Returns a Bearer JWT valid for 24 hours. Include the token in subsequent requests as `Authorization: Bearer <token>`.

REQUEST BODY SCHEMA: application/x-www-form-urlencoded
required

grant_type >	Grant Type (string) or Grant Type (null) (Grant Type)
username required	string (Username)
password required	string <password> (Password)
scope	string (Scope) Default: `""`
client_id >	Client Id (string) or Client Id (null) (Client Id)

client_secret > Client Secret (string) or Client Secret (null) <password> (Client Secret)

Responses

> **200** JWT access token

— **401** Username or email not found, or password incorrect

> **422** Validation Error

POST /api/v1/auth/login

Response samples

200 422

Content type
application/json

Copy


```
{
    "access_token": "eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9.eyJzdWIiOiJqYW5lX2R
    "token_type": "bearer"
}
```


prices

Create a price entry

Record a new food price observation. Requires authentication. The `currency_code` must exist in the currencies reference table.

AUTHORIZATIONS: > *OAuth2PasswordBearer*

REQUEST BODY SCHEMA: application/json
required

`date` required	string <date> (Date)
`countryiso3` required	string (Countryiso3) = 3 characters
`admin1` >	Admin1 (string) or Admin1 (null) (Admin1)
`admin2` >	Admin2 (string) or Admin2 (null) (Admin2)
`market_id` required	integer (Market Id)
`commodity_id` required	integer (Commodity Id)
`category` >	Category (string) or Category (null) (Category)
`unit` >	Unit (string) or Unit (null) (Unit)
`priceflag` >	Priceflag (string) or Priceflag (null) (Priceflag)
`pricetype` >	Pricetype (string) or Pricetype (null) (Pricetype)
`currency_code` required	string (Currency Code) [1 .. 10] characters
`price` >	Price (number) or Price (null) (Price)
`usdprice` >	Usdprice (number) or Usdprice (null) (Usdprice)

Responses

> **201** The newly created price record

— **422** Unknown currency code

POST /api/v1/prices

Request samples

Payload

Content type
application/json

Copy


```
{
    "admin1": "Oromia",
    "admin2": "Jimma",
    "category": "cereals and tubers",
    "commodity_id": 10,
    "countryiso3": "ETH",
    "currency_code": "ETB",
    "date": "2024-01-15",
    "market_id": 1,
    "price": 45.5,
    "priceflag": "actual",
    "pricetype": "Retail",
    "unit": "KG",
    "usdprice": 0.82
}
```


Response samples

201

Content type
application/json

Copy


```
{
    "admin1": "Oromia",
    "admin2": "Jimma",
    "category": "cereals and tubers",
    "commodity_id": 10,
    "countryiso3": "ETH",
    "created_at": "2024-01-15T12:00:00Z",
    "currency_code": "ETB",
    "date": "2024-01-15",
```

```
    "id": 1001,
    "market_id": 1,
    "price": 45.5,
    "priceflag": "actual",
    "pricetype": "Retail",
    "unit": "KG",
    "updated_at": "2024-01-15T12:00:00Z",
    "usdprice": 0.82
  }
```


List price records

Return a paginated list of price observations. Filter by country (ISO 3166-1 alpha-3), commodity, market, and/or date range. Results are ordered by date descending. Maximum page size is 200.

QUERY PARAMETERS

- `country` ›
  Country (string) or Country (null) (Country)
  ISO 3166-1 alpha-3 country code, e.g. `ETH`
- `commodity_id` ›
  Commodity Id (integer) or Commodity Id (null) (Commodity Id)
  Filter by commodity ID
- `market_id` ›
  Market Id (integer) or Market Id (null) (Market Id)
  Filter by market ID
- `date_from` ›
  Date From (string) or Date From (null) (Date From)
  Inclusive start date (YYYY-MM-DD)
- `date_to` ›
  Date To (string) or Date To (null) (Date To)
  Inclusive end date (YYYY-MM-DD)
- `page`
  integer (Page) `>= 1`
  Default: `1`
  Page number (1-indexed)
- `page_size`
  integer (Page Size) `[ 1 .. 200 ]`
  Default: `50`
  Number of records per page

Responses

> **200** Paginated list of price records with total count

> **422** Validation Error

GET /api/v1/prices

Response samples

200 422

Content type
application/json

Copy Expand all Collapse all


```
{
  - "items": [
      + { … }
    ],
    "page": 1,
    "page_size": 50,
    "total": 4823
}
```


Get a price record by ID

PATH PARAMETERS

price_id required	integer (Price Id)

Responses

> **200** The requested price record

— **404** Price not found

> **422** Validation Error

GET /api/v1/prices/{price_id}

Response samples

200 422

Content type
application/json

Copy


```
{
    "admin1": "Oromia",
    "admin2": "Jimma",
    "category": "cereals and tubers",
    "commodity_id": 10,
    "countryiso3": "ETH",
    "created_at": "2024-01-15T12:00:00Z",
    "currency_code": "ETB",
    "date": "2024-01-15",
    "id": 1001,
    "market_id": 1,
    "price": 45.5,
    "priceflag": "actual",
    "pricetype": "Retail",
    "unit": "KG",
    "updated_at": "2024-01-15T12:00:00Z",
    "usdprice": 0.82
}
```


Update a price record

Partially update a price record. Only fields included in the request body are changed. Requires authentication.

AUTHORIZATIONS: > *OAuth2PasswordBearer*

PATH PARAMETERS

price_id required	integer (Price Id)

REQUEST BODY SCHEMA: application/json
required

date >	Date (string) or Date (null) (Date)
countryiso3 >	Countryiso3 (string) or Countryiso3 (null) (Countryiso3)
admin1 >	Admin1 (string) or Admin1 (null) (Admin1)
admin2 >	Admin2 (string) or Admin2 (null) (Admin2)
market_id >	Market Id (integer) or Market Id (null) (Market Id)
commodity_id >	Commodity Id (integer) or Commodity Id (null) (Commodity Id)
category >	Category (string) or Category (null) (Category)
unit >	Unit (string) or Unit (null) (Unit)
priceflag >	Priceflag (string) or Priceflag (null) (Priceflag)
pricetype >	Pricetype (string) or Pricetype (null) (Pricetype)
currency_code >	Currency Code (string) or Currency Code (null) (Currency Code)
price >	Price (number) or Price (null) (Price)
usdprice >	Usdprice (number) or Usdprice (null) (Usdprice)

Responses

> **200** The updated price record

— **404** Price not found

— **422** Unknown currency code

PUT /api/v1/prices/{price_id}

Request samples

Payload

Content type
application/json

Copy

```
{
    "price": 48,
    "usdprice": 0.86
}
```

Response samples

200

Content type
application/json

Copy

```
{
    "admin1": "Oromia",
    "admin2": "Jimma",
    "category": "cereals and tubers",
    "commodity_id": 10,
    "countryiso3": "ETH",
    "created_at": "2024-01-15T12:00:00Z",
    "currency_code": "ETB",
    "date": "2024-01-15",
    "id": 1001,
    "market_id": 1,
    "price": 45.5,
    "priceflag": "actual",
    "pricetype": "Retail",
    "unit": "KG",
    "updated_at": "2024-01-15T12:00:00Z",
    "usdprice": 0.82
}
```


Delete a price record

Permanently delete a price record by ID. Requires authentication.

AUTHORIZATIONS: > *OAuth2PasswordBearer*

PATH PARAMETERS

price_id required	integer (Price Id)

Responses

— **204** Successful Response

— **404** Price not found

> **422** Validation Error

DELETE /api/v1/prices/{price_id}

Response samples

422

Content type
application/json

Copy Expand all Collapse all

```
{
  - "detail": [
      + { … }
    ]
}
```

analytics

Monthly price trend for a commodity in a country

Return monthly average USD prices for the specified country and commodity. Optionally narrow the window with `date_from` / `date_to`. Returns 404 if no data exists for the combination.

QUERY PARAMETERS

`country` required	string (Country) ISO 3166-1 alpha-3 country code, e.g. `ETH`
`commodity_id` required	integer (Commodity Id) Commodity ID from `/commodities`
`date_from` >	Date From (string) or Date From (null) (Date From) Inclusive start date (YYYY-MM-DD)
`date_to` >	Date To (string) or Date To (null) (Date To) Inclusive end date (YYYY-MM-DD)

Responses

> **200** Time series of monthly average prices

— **404** No data found for the given country code

> **422** Validation Error

GET /api/v1/analytics/trends

Response samples

200 422

Content type
application/json

Copy Expand all Collapse all

```
{
    "commodity_id": 10,
    "country": "ETH",
  - "points": [
      + { … },
      + { … },
      + { … }
    ]
}
```

Price volatility ranking for a country

Return the top-N commodities ranked by coefficient of variation (CV = stddev / mean) of their USD price in the given country. Higher CV indicates more volatile pricing.

QUERY PARAMETERS

`country` string (Country)

required

ISO 3166-1 alpha-3 country code, e.g. `ETH`

limit

integer (Limit) [1 .. 50]

Default: `10`

Maximum number of commodities to return

Responses

> **200** List of commodities sorted by volatility (highest first)

— **404** No data found for the given country code

> **422** Validation Error

GET /api/v1/analytics/volatility

Response samples

200 422

Content type

application/json

Copy Expand all Collapse all

```
{
    "country": "ETH",
  - "items": [
      + { … },
      + { … }
    ]
}
```

Compare commodity prices across countries

Return the average USD price of a single commodity in every country that has recorded it, sorted highest-to-lowest. Useful for spotting regional price disparities.

QUERY PARAMETERS

`commodity_id` required	integer (Commodity Id) Commodity ID from `/commodities`
`date_from` >	Date From (string) or Date From (null) (Date From) Inclusive start date (YYYY-MM-DD)
`date_to` >	Date To (string) or Date To (null) (Date To) Inclusive end date (YYYY-MM-DD)

Responses

> **200** Average USD price per country, sorted descending

> **422** Validation Error

GET `/api/v1/analytics/regional-comparison`

Response samples

200 | 422

Content type
application/json

Copy Expand all Collapse all


```
{
    "commodity_id": 10,
  - "countries": [
      + { … },
      + { … },
      + { … }
    ]
}
```

Food crisis scores for all countries

Compute a composite food-crisis score (0–100) for every country using a 12-month trailing window. The score blends volatility (40%), price trend (35%), and commodity breadth (25%). Severity bands: **stable** < 25 ≤ **moderate** < 50 ≤ **high** < 75 ≤ **critical**.

Responses

> **200** Crisis scores for all countries with data, sorted by score descending

GET /api/v1/analytics/crisis-scores

Response samples

200

Content type
application/json

Copy Expand all Collapse all


```
[
 - {
      "breadth_score": 55.9,
      "countryiso3": "SOM",
      "crisis_score": 72.4,
      "severity": "high",
      "trend_score": 81.3,
      "volatility_score": 68.1
   }
]
```


Food crisis score for a single country

Return the composite food-crisis score for the given ISO 3166-1 alpha-3 country code.

PATH PARAMETERS

country required	string (Country)

Responses

> **200** Crisis score breakdown for the requested country

— **404** No data found for the given country code

> **422** Validation Error

GET /api/v1/analytics/crisis-scores/{country}

Response samples

200 422

Content type
application/json

Copy

```
{
    "breadth_score": 55.9,
    "countryiso3": "SOM",
    "crisis_score": 72.4,
    "severity": "high",
    "trend_score": 81.3,
    "volatility_score": 68.1
}
```

Market summary with latest prices

Return aggregate statistics and the most recent price for each commodity recorded at the specified market.

PATH PARAMETERS

market_id required	integer (Market Id)

Responses

> **200** Market metadata, date range, and latest price per commodity

— **404** Market not found

> **422** Validation Error

GET /api/v1/analytics/markets/{market_id}/summary

Response samples

200 422

Content type
application/json

Copy Expand all Collapse all


```
{
    "commodity_count": 8,
    "countryiso3": "ETH",
    "date_from": "2019-01-15",
    "date_to": "2024-01-15",
  - "latest_prices": [
      + { … },
      + { … }
    ],
    "market_id": 1,
    "market_name": "Addis Ababa",
    "total_prices": 1240
}
```

reference

List countries with price data

Return all countries that have at least one price record, ordered by record count descending.

Responses

> **200** List of ISO 3166-1 alpha-3 country codes with their price record counts

GET /api/v1/countries

Response samples

200

Content type
application/json

Copy Expand all Collapse all


```
[
 - {
      "count": 4823,
      "countryiso3": "ETH"
   }
]
```


List all commodities

Return all food commodities tracked in the system, ordered alphabetically by name.

Responses

> **200** List of commodities with ID, category, and name

GET /api/v1/commodities

Response samples

200

Content type
application/json

Copy Expand all Collapse all


```
[
 - {
      "category": "cereals and tubers",
      "id": 10,
      "name": "Maize"
   }
]
```


List all currencies

Return all currency codes used in price records, ordered by currency code.

Responses

> **200** List of currency codes and their full names

GET /api/v1/currencies

Response samples

200

Content type
application/json

Copy Expand all Collapse all


```
[
 - {
      "code": "ETB",
      "name": "Ethiopian Birr"
   }
]
```


List markets

Return all markets, optionally filtered by country. Results are ordered alphabetically by market name.

QUERY PARAMETERS

`country` >

Country (string) or Country (null) (Country)
Filter by ISO 3166-1 alpha-3 country code, e.g. `ETH`

Responses

> **200** List of markets with location details

> **422** Validation Error

GET /api/v1/markets

Response samples

200 422

Content type
application/json

Copy Expand all Collapse all


```
[
 - {
      "admin1": "Addis Ababa",
      "countryiso3": "ETH",
      "id": 1,
      "latitude": 9.025,
      "longitude": 38.747,
      "name": "Addis Ababa"
   }
]
```